

Peel and Plate Fruit SOP

One episode peels one banana on the board in the middle of the table, throws the peel away, cuts the flesh into four segments with a knife on the plate to the right of the board, arranges the segments into one row, and wipes the board.

The episode starts with the banana lying in its start zone. The order never changes: peel it, plate the flesh, throw the peel away, cut, arrange, wipe. Nothing is cut until every peel strip is off the flesh and the peel is in the bin, and the board is not wiped until the row is arranged.

The **board** in the middle of the table is where the whole peel happens. The **plate** sits to the right of the board and stays there for the whole episode. It is never carried, and the banana never goes onto it until it is peeled.

The banana lies with its **banana stem** pointing left and its **banana tip** pointing right. The left gripper closes on the banana stem and lifts the stem end a little clear of the board while the right gripper peels, drawing every peel strip to the left. The right gripper does everything else: it starts every peel strip at the stem and pulls it, plates the flesh, cuts the segments with the knife, arranges them, and wipes the board. The left gripper also rolls the banana in place between pulls and clears the peel, which goes straight left into the waste bin.

The banana is peeled lying on the board, never held up in the air. The left gripper lifts the stem end a little clear of the board, but the body of the banana stays down on the board. The banana is turned by rolling it in place on the board, never by lifting or twisting it in a gripper. Peeled fruit and banana segments rest on the plate face and nowhere else.

The table is set up in one of three ways. Only the banana moves; the board, the plate, the knife rest, the cloth rest, and the waste bin are in the same place in all three.

- **Config L:** the banana lies at the back-left, on the bare table left of the board.
- **Config M:** the banana lies on the board, in the middle of the table.
- **Config R:** the banana lies at the back-right, on the bare table behind the plate and the knife rest.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the banana starts at the back-left (**Config L**), on the board (**Config M**), or at the back-right (**Config R**), always with its banana stem pointing left. One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the banana's side lifts it onto the board — the **left gripper** in Config L, the **right gripper** in Config R. In Config M it is already there and nothing touches it

until the stem hold. No arm reaches across the table.

- **Fixed roles:** from the first peel pull on, the right gripper peels, plates, cuts, arranges, and wipes, and the left gripper holds the stem, rolls the banana, and clears the peel, in every config. The board, the plate, the knife rest, the cloth rest, and the waste bin never move.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the board in the middle, the banana in the start zone for this episode's config, the plate spot right of the board, the knife rest right of the plate, the cloth rest at the front right, and the waste bin at the front left corner.
3. The board is visible from above and is not cut off at the edge of frame, so every peel strip can be seen coming off and any pulp left on the board can be seen.
4. The plate is visible from above, so every cut and the finished row can be seen.
5. Both arms are at home with grippers open.
6. The table is clean, dry, and bare apart from the board, the banana, the plate, the knife, the cloth, and the waste bin.
7. The board does not slide when a gripper presses on it, and the plate does not slide when the knife presses on it.
8. The right arm reaches the whole board, the plate face, the knife rest, the cloth rest, and (Config R) the banana at the back-right. The left arm reaches the whole board, the waste bin, and (Config L) the banana at the back-left.
9. The right gripper reaches the plate face without stretching across the body. If it cannot, move the plate spot nearer the board before the session starts.
10. The board is wide enough to take the banana lying across it with its three peel strips open to the left of it, without any of them hanging off the edge.

Materials checklist

1. One whole **banana** lies flat in the start zone for this episode's config, its **banana stem** pointing left and the banana tip pointing right:
 - **Config L:** back-left, on the bare table left of the board and behind the waste bin.

- **Config M:** centered on the board, clear of its edges, with room to the left for the peel strips to open flat.
 - **Config R:** back-right, on the bare table behind the plate and the knife rest.
2. The banana is ripe and unsplit with its **banana stem** intact, firm enough that the flesh does not break apart under a correct peel.
 3. One flat **board** lies in the middle of the table, clean and dry, holding nothing but the banana in Config M and nothing at all in Config L and R.
 4. One clean dry **plate** lies flat and face up on the plate spot, right of the board. The same size of plate is used every episode.
 5. One clean dry **knife** lies flat on the knife rest, right of the plate, with its **knife handle** toward the board and its **knife blade** pointing away from both arms.
 6. One folded dry **cloth** lies on the cloth rest at the front right. The cloth goes back to the cloth rest at the end of the wipe and is never thrown away.
 7. One open **waste bin**, empty, at the front left corner.
 8. The plate face is clear, and the two start zones not used by this episode's config are bare.

Workspace layout

- **Start zone:** where the banana lies when recording starts — back-left (**Config L**), on the board (**Config M**), or back-right (**Config R**).
- **Board:** the middle of the table. The banana is peeled here, and in Config M it starts here. Nothing else is worked on it.
- **Plate spot:** right of the board, the plate. The plate lies here for the whole episode and is never picked up or moved. All cutting and arranging happens on it.
- **Knife rest:** right of the plate, where the knife lies when it is not cutting.
- **Cloth rest:** front right, the folded cloth.
- **Waste bin:** front left corner, open and empty. The peel goes straight into it off the board.

The plate, the knife, and the cloth are on the right, where the right gripper works. The bin is on the left, where the left gripper takes the peel. The banana starts in the start zone for the episode's config and is peeled on the board.

Vocabulary

Objects

- **Banana stem:** the short stalk at the end of the banana where it was cut from the bunch. It points left. Every peel strip starts here and the left gripper holds here. The **banana tip** is the end opposite it, pointing right.
- **Banana skin:** the outer covering of the banana, taken off as three peel strips. The **banana peel** is all of that skin once it is off the flesh, still joined at the banana tip.

- **Peel strip:** one length of banana skin drawn away from the flesh in one motion.
- **Peeled fruit:** banana flesh with no skin and no peel strip left on it.
- **Banana segment:** one length of banana flesh cut off the peeled fruit with the knife. Segments are cut to an even size, four from one banana.
- **Board:** the flat board in the middle of the table where the banana is peeled.
- **Start zone:** where the banana lies at the start of the episode — back-left (**Config L**), on the board (**Config M**), or back-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Plate:** the clean plate on the plate spot. Its **plate rim** is the raised outer edge. Its **plate face** is the flat inner surface the fruit sits on.
- **Knife:** the knife used to cut the peeled fruit into segments. Its **knife handle** is the end the right gripper closes on. Its **knife blade** is the cutting edge. The knife lives on the knife rest and only ever travels between the knife rest and the plate.
- **Segment row:** the single row of four banana segments running left to right across the middle of the plate face, parallel to the front edge.
- **Arranged plate:** all four banana segments in one segment row, none stacked, none touching the plate rim.
- **Cloth:** the folded cloth used to wipe the board, then laid back flat on the cloth rest. Its **cloth face** is the flat side that touches the board.
- **Debris:** any banana segment, peel strip, or pulp outside the plate and the waste bin.

Actions

- **Stem hold:** the left gripper closed on the banana stem, lifting the stem end a little clear of the board and holding it there while the right gripper works on it. The body of the banana stays lying on the board and never leaves it.
- **Board hold:** the left gripper closed on the banana or the banana peel and holding it down against the board while the right gripper works on it. The thing being held never leaves the surface.
- **Peel pull:** the right gripper closed on the banana skin at the banana stem and drawing it to the left, past the stem end, in one motion. The skin opens at the stem, the strip folds back on itself as it goes, and it comes free along the whole length of the banana, staying joined at the banana tip.
- **Roll in place:** the left gripper turning the banana about its own long axis by pressing its side while it stays lying flat on the board, so the side that was underneath comes up and the next length of banana skin can be reached. The banana stem stays pointing left, and the banana is never lifted or twisted in a gripper to turn it.
- **Cut:** the right gripper holding the knife by its handle, lowering the knife blade square across the peeled fruit, pressing straight down until the blade meets the plate face, then lifting the blade straight up. One cut in one press, never a sawing or sideways drag.

- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Wipe pass:** one continuous draw of the cloth face from left to right across the board.
- **Release point:** over the plate face, over the board, over the knife rest, over the cloth rest, or over the waste bin. A gripper opens nowhere else.

Arm assignments

- **Right gripper:** in Config R lifts the banana from the back-right onto the board; then starts every peel strip at the banana stem and pulls it to the left, takes the peeled fruit at the middle of its length and moves it to the plate, takes the knife, cuts the four banana segments, returns the knife to the knife rest, arranges the segment row, takes the cloth, wipes the board, and parks the cloth.
- **Left gripper:** in Config L lifts the banana from the back-left onto the board; then takes the stem hold on the banana, lifting the stem end a little, while every peel strip is pulled, rolls the banana in place between pulls, holds the peel down on the board while the flesh comes free of it, and takes the peel straight off the board into the waste bin.

Neither gripper takes the other's jobs. The left gripper never peels, never cuts, and never touches the knife or the cloth. The right gripper never takes the stem hold and never handles the peel once it is off the flesh. The left gripper stays clear of the plate for the whole episode.

Steps

Only Step 1 depends on the config: in Config L the **left gripper** lifts the banana from the back-left onto the board, in Config R the **right gripper** lifts it from the back-right onto the board, and in Config M it is already on the board. From the first peel pull on, every line is the same in all three. Steps 1 to 5 are one plate. Step 6 ends the episode.

Step 1: Peel the banana

Goal: the banana flesh peeled clean, lying in its open peel on the board.

Look where the banana is before the first reach.

- **IF the banana is at the back-left (Config L):** the **left gripper** closes on the banana around the middle of its length, **lifts** it straight up, carries it level to the **board**, lowers it flat and centered on the board with the banana stem still pointing left, and releases.
- **IF the banana is on the board (Config M):** nothing brings it there and nothing moves it. Both grippers go straight to 1.1.
- **IF the banana is at the back-right (Config R):** the **right gripper** closes on the banana around the middle of its length, **lifts** it straight up, carries it level to the **board**, lowers it flat and centered on the board with the banana stem still pointing left, and releases.

Then, in all three, the banana lies flat and centered on the board with its banana stem pointing left, and the peel runs the same way.

1.1 Pull the first peel strip

- The **left gripper** closes on the **banana stem** and takes its **stem hold**, lifting the stem end a little clear of the board while the body of the banana stays lying on it, and holds it there for the whole pull.
- The **right gripper** closes on the **banana skin** at the banana stem and makes one **peel pull**, drawing it to the left, past the stem end, so the skin opens at the stem and the strip comes free the whole length of the banana.
- The **right gripper** lays the peel strip open flat on the board to the left of the banana and releases it.

1.2 Turn the banana and pull the rest

- The **left gripper** releases the banana stem and the banana rests flat on the board.
- The **left gripper** does a **roll in place**, pressing the side of the banana until the part still covered in skin faces up.
- The **left gripper** takes the **stem hold** again.
- The **right gripper** makes the next **peel pull** on the next edge of the banana skin the same way and lays it open flat to the left.
- Repeat the turn and the pull until three peel strips lie open to the left of the flesh and no skin is left on the **banana flesh**.
- If a peel strip tears before it is free, take the torn end once and pull it the rest of the way.

The banana stays flat on the board for the whole peel. Never lift the banana into the air to peel it, never carry it over to the plate while skin is still on it, and never set the bare flesh down on the board or the table. Every pull travels left. To reach a side that is turned away, the left gripper lets go, rolls the banana in place, and takes the stem hold again.

Check: no banana skin and no peel strip is left on the flesh, the three peel strips lie open flat on the board to the left still joined at the banana tip, and the banana is still on the board.

Step 2: Plate the peeled fruit and clear the peel

Goal: the peeled fruit lying on the plate face, and the banana peel in the waste bin.

2.1 Move the peeled fruit to the plate

- The **left gripper** keeps its hold so the peel stays down on the board.
- The **right gripper** closes on the **peeled fruit** around the middle of its length, lifts it straight up clear of the open peel, carries it level to the **plate**, lowers it onto the **plate face** with its length running left to right, and releases.
- The **left gripper** releases the peel.

2.2 Discard the banana peel

- The **left gripper** closes on the **banana peel** at the banana tip, lifts it straight up off the board, carries it to the **waste bin**, and releases it over the bin.
- If the peel came off as separate peel strips, the **left gripper** takes each one off the board the same way and releases it over the waste bin, until no banana skin is left on the board.

The peel goes straight from the board into the waste bin as soon as the flesh is plated. It is never set down anywhere on the way, and it never goes right, over or onto the plate.

Check: the peeled fruit lies on the plate face with no skin on it, the waste bin holds the banana peel, no peel is on the board, on the table, on the plate, or outside the bin, and the board is clear of everything but pulp.

Step 3: Cut the fruit into segments

Goal: the peeled fruit cut into four even banana segments, all on the plate face.

3.1 Take the knife

- The **right gripper** closes on the **knife handle** on the knife rest, lifts the knife clear, and brings it over the **plate face** with the **knife blade** square across the peeled fruit.
- The knife travels only between the knife rest and the plate. It never passes over the board, the waste bin, and the knife blade never points at the other arm.

3.2 Cut

- The **right gripper** makes the first **cut** about a quarter of the length in from the banana stem end, pressing straight down until the knife blade meets the plate face, then lifting straight up.
- Make the second cut the same distance again toward the banana tip, and the third the same distance after that, so the fruit ends as four even **banana segments**.
- Space the cuts by eye so the four segments come out even.
- Nothing steadies the fruit. The **left gripper** stays clear of the plate for the whole of Step 3.

Every cut is one straight press down onto the plate face, never a sawing or sideways drag. If a piece rolls out of place or off the plate face, return the knife to the knife rest first, then set the piece back on the plate face with the right gripper before cutting again.

3.3 Return the knife

- The **right gripper** carries the knife to the **knife rest**, lays it down flat with the knife handle toward the board and the knife blade pointing away from both arms, and releases.

The knife goes back on its rest before anything else is picked up. Nothing is arranged, cleared, or wiped with the knife in the gripper.

Check: four even banana segments sit on the plate face, cut all the way through and none crushed or torn, the knife lies flat on the knife rest, and no banana is on the board or the table.

Step 4: Arrange the segments on the plate

Goal: an arranged plate, one row of four banana segments with none stacked and none touching the plate rim.

- The **right gripper** closes on one **banana segment** and places it across the middle of the **plate face** in one **segment row** running left to right, beside the segment next to it and clear of the plate rim.
- Repeat until all four banana segments sit in one segment row.
- Release once every segment is resting.

No segment sits on top of another, hangs over the plate rim, or leaves a visible gap in the row.

Check: four banana segments sit in one segment row across the middle of the plate face, none are stacked, none touch or overhang the plate rim, and the row reads as a row with no gap.

Step 5: Wipe the board

Goal: the whole board clean and dry, with no debris and no pulp on it, and the cloth back on the cloth rest.

5.1 Pick up the cloth

- The **right gripper** closes on the folded **cloth** on the cloth rest, carries it to the far edge of the board on a path clear of the plate, and lowers it until the **cloth face** lies flat on the board.

5.2 Wipe

- Start each pass at the left edge of the board.
- Draw the cloth from left to right in one continuous **wipe pass**, keeping the cloth face flat on the board for the whole pass.
- Lift at the right edge of the board, never carrying the cloth on over the plate.
- Return to the left edge of the board through the air, and start the next wipe pass nearer the front edge so it overlaps the pass just made.
- Repeat, working from the far edge to the near edge, until the last wipe pass reaches the near edge of the board.
- If any debris or pulp is left after the last wipe pass, run one short repeat pass over that spot.

5.3 Park the cloth

- The **right gripper** carries the cloth to the **cloth rest**, lays it down flat, releases, and lifts clear.

The cloth is not thrown away. It goes back on the cloth rest at the end of every wipe.

Every wipe pass runs left to right and parallel to the front edge. The cloth stays over the board and never passes over the plate.

Check: the board is clean and dry with no debris and no pulp, the cloth lies flat on the cloth rest, and the arranged plate is untouched with the row intact.

Step 6: Confirm the plate and end the episode

- Confirm the plate spot holds one arranged plate with the row intact, the knife lies flat on the knife rest, the board is clean and dry and in its place, the cloth lies flat on the cloth rest, and the waste bin holds the peel.
- Return both arms home, clear of the plate and the board, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Tip the banana segments off the plate into the waste bin.
2. Wash and dry the plate, then lay it flat and face up on the plate spot, right of the board.
3. Wash and dry the knife and lay it flat on the knife rest, right of the plate, with the knife handle toward the board and the knife blade pointing away from both arms.
4. Shake out the cloth, fold it flat and dry, and lay it back on the cloth rest at the front right. Replace a cloth that is wet, torn, or shedding fibre.
5. Empty the waste bin and set it back at the front left corner, open.
6. Wipe any debris and pulp off the board and the table and confirm both are clean and dry. A wet board lets the banana slide under a stem hold.
7. Confirm the board is square in the middle of the table and does not slide, and that the plate face is clear.
8. Lay one whole banana flat, with its banana stem pointing left and the banana tip pointing right, in the start zone for the next episode's config — back-left (Config L), centered on the clean board (Config M), or back-right (Config R). The other two zones stay bare.
9. Replace a banana that is bruised, split, or so ripe that the flesh breaks apart.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config M (the banana already lying on the board). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the banana is not in the start zone for the config; a gripper reaches across the table for the banana; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the banana starts at the back-left, on the board, or at the back-right; the gripper on its side lifts it onto the board — the left in Config L, the right in Config R, neither in Config M; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the banana is before the first reach, then follow that config's IF line through Step 1.

Violation: Wrong order

- **Visible cue:** the flesh is cut before it is fully peeled, the knife comes out before the peel is in the bin, or the board is wiped before the row is arranged.
- **SOP rule broken:** Steps 1 to 5, the order never changes, the flesh is fully peeled before any cutting, and the peel is binned before the knife is taken.
- **Coaching note:** peel, plate, discard, cut, arrange, wipe. Check the flesh for leftover skin before you cut it.

Violation: Arms swapped roles

- **Visible cue:** the left gripper starts or pulls a peel strip, moves the flesh to the plate, takes the knife, arranges the row, or takes the cloth; or the right gripper takes the stem hold, rolls the banana, or carries the peel to the waste bin.
- **SOP rule broken:** Steps 1 to 5, the right gripper does the peel, the knife, the arranging, and the wipe; the left gripper holds the stem, rolls the banana, and clears the peel.
- **Coaching note:** right pulls, plates, cuts, and wipes. Left holds the stem, rolls the banana, and takes the peel away.

Violation: Banana peeled off the board

- **Visible cue:** the banana is lifted into the air to be peeled, the body of the banana comes off the board during a pull, or it is taken off the board and peeled on the bare table instead.
- **SOP rule broken:** Step 1, the banana lies flat on the board for the whole peel and only the stem end is lifted a little clear of it.

- **Coaching note:** hold it up by the stem and work against the board. The banana stays on the board from the first strip to the last.

Violation: Banana stem not held during a pull

- **Visible cue:** the left gripper is off the banana while a peel strip is pulled, the stem end is not lifted a little clear of the board for a pull, or the body of the banana slides, rolls, or lifts off the board during a pull. A left gripper off the banana for a roll in place between pulls is not this violation, as long as the stem hold is back on before the next pull starts.
- **SOP rule broken:** Steps 1.1 and 1.2, the left gripper holds the stem end up a little for the whole of every pull and releases only for a roll in place.
- **Coaching note:** the hold goes on before the right gripper starts and comes off after it finishes. Turn the banana between pulls, never during one.

Violation: Peel strip started or pulled wrong

- **Visible cue:** the banana skin opens along the side of the banana or at the banana tip, or the flesh is squeezed out through the skin, instead of opening at the banana stem; or a peel strip is drawn to the right toward the banana tip instead of to the left past the stem end.
- **SOP rule broken:** Step 1.1, every peel strip is started at the banana stem and drawn to the left, past the stem end.
- **Coaching note:** every strip starts at the stem only, and every pull travels left.

Violation: Banana turned wrong

- **Visible cue:** the banana is lifted, twisted in a gripper, or dragged off the board to bring the unpeeled side up, instead of being rolled in place on the board by the left gripper; or it is turned end for end so the banana stem no longer points left.
- **SOP rule broken:** Step 1.2, the banana is turned by a roll in place with the banana stem still pointing left.
- **Coaching note:** let go, press the side with the same gripper to roll it on the board, then take the stem hold again.

Violation: Flesh plated with skin still on it

- **Visible cue:** the flesh is carried to the plate with a peel strip or a patch of banana skin still on it, or a peel strip is still being pulled after the flesh has reached the plate.
- **SOP rule broken:** Steps 1.2 and 2.1, no skin and no peel strip is left on the flesh before it leaves the board.
- **Coaching note:** look the whole length over before you lift it. Every strip comes off on the board.

Violation: Peel or peel strip taken to the plate side

- **Visible cue:** the banana peel or an open peel strip is carried over the plate, laid on the plate, or set down on the plate spot or the knife rest; or the peel is carried right instead of left to the waste bin.
- **SOP rule broken:** Steps 1 and 2.2, the peel stays on the board and then goes left, straight into the waste bin.
- **Coaching note:** the peel only ever moves left. Nothing off the skin goes near the plate.

Violation: Peeled fruit or a banana segment off the plate face

- **Visible cue:** the peeled fruit or a banana segment is set down on the board, on the table, on the knife rest, or on the plate rim.
- **SOP rule broken:** Steps 2.1, 3, and 4, peeled fruit and banana segments rest on the plate face and nowhere else.
- **Coaching note:** flesh goes straight onto the plate face and stays there.

Violation: Knife taken or carried wrong

- **Visible cue:** the knife is taken anywhere but the knife handle, carried over the board, the peel rest, or the waste bin, held with the knife blade pointing at the other arm, or dragged across the table instead of lifted clear.
- **SOP rule broken:** Steps 3.1 and 3.3, the knife is held by the handle and travels only between the knife rest and the plate.
- **Coaching note:** handle only, lifted clear, and straight between the rest and the plate.

Violation: Segments not cut cleanly

- **Visible cue:** at the end of Step 3 the flesh is in fewer or more than four segments or the segments are visibly uneven, a cut does not go all the way through and leaves two segments joined, or the flesh is crushed, torn, or sawed apart instead of cut in one straight press.
- **SOP rule broken:** Step 3.2, three cuts, each one straight press down to the plate face, giving four even banana segments.
- **Coaching note:** square the blade across the fruit, press straight down to the plate, and take the same distance in for every cut.

Violation: Fruit steadied during a cut

- **Visible cue:** the left gripper touches the peeled fruit or a banana segment, comes over the plate, or steadies anything while the knife is over the plate.
- **SOP rule broken:** Step 3.2, nothing steadies the fruit and the left gripper stays clear of the plate for the whole of Step 3.
- **Coaching note:** the cut is one clean press. If a piece rolls, put the knife back first, then set the piece straight.

Violation: Knife not returned to its rest

- **Visible cue:** the knife is still in the gripper when the row is arranged, the peel is cleared, or the board is wiped; or at the end of the episode it is on the plate, on the board, on the table, or dropped short of the knife rest.
- **SOP rule broken:** Step 3.3, the knife goes back flat on the knife rest before anything else is picked up.
- **Coaching note:** knife down first, then everything else.

Violation: Row not arranged

- **Visible cue:** at the end of Step 4 a banana segment sits on top of another, touches or overhangs the plate rim, or the row has a visible gap in it.
- **SOP rule broken:** Step 4, one segment row across the middle of the plate face, each segment beside the one next to it, clear of the plate rim, and none stacked.
- **Coaching note:** set each segment down beside its neighbor, not on it, and keep the whole row clear of the rim.

Violation: Peel not discarded

- **Visible cue:** the cutting starts with the banana peel or a peel strip still on the board, the peel is set down anywhere on its way to the bin, or at the end of Step 2 it is still on the board, on the table, or on the plate, or dropped outside the waste bin.
- **SOP rule broken:** Step 2.2, the peel goes straight off the board into the waste bin as soon as the flesh is plated.
- **Coaching note:** peel off the board and into the bin before the knife comes out. Nothing is set down in between.

Violation: Cloth taken over the plate

- **Visible cue:** the cloth is carried over the arranged plate on its way to or from the board, drawn out past the right edge of the board toward the plate, or used to wipe the plate or the plate spot.
- **SOP rule broken:** Steps 5.1 and 5.2, the cloth travels clear of the plate and every wipe pass stops at the right edge of the board.
- **Coaching note:** the cloth only ever touches the board. A cloth near an arranged plate breaks the row.

Violation: Wipe skipped, misdirected, or the board left dirty

- **Visible cue:** Step 5 is skipped, debris or pulp is still on the board at the end of it and the episode moves on without a repeat pass, the wipe passes run right to left, front to back, or on the diagonal, or a strip of the board is left untouched between two passes.

- **SOP rule broken:** Steps 5 and 5.2, left to right wipe passes from the far edge to the near edge, each overlapping the pass just made, until the board is clean and dry.
- **Coaching note:** keep every pass parallel to the front edge, work far to near, overlap the last one, then check and run a short repeat pass over any spot still dirty.

Violation: Cloth not parked on the cloth rest

- **Visible cue:** at the end of Step 5 the cloth is left on the board, on the table, or on the plate, dropped short of the cloth rest, draped over the edge of it, or put in the waste bin.
- **SOP rule broken:** Step 5.3, the cloth is laid flat back on the cloth rest at the end of the wipe and is never thrown away.
- **Coaching note:** the cloth lives on its rest. Lay it flat there, do not bin it.

Violation: Plate moved off its spot

- **Visible cue:** either gripper picks up, slides, spins, or nudges the plate, or the plate ends the episode anywhere other than the plate spot.
- **SOP rule broken:** Steps 1 to 5, the plate lies on the plate spot for the whole episode and is never picked up or moved.
- **Coaching note:** bring the fruit to the plate. The plate itself never moves.

Violation: Board shifted out of place

- **Visible cue:** the board slides, spins, or is pushed out of the middle of the table during the a pull, a roll, or the wipe, and is not squared up before the next action; or the board is lifted.
- **SOP rule broken:** Steps 1 to 5, the board stays square in the middle of the table and is worked on, never moved.
- **Coaching note:** press down into the board, not sideways across it. If it shifts, stop and square it before going on.

Violation: Bad pick, carry, or release

- **Visible cue:** a thing stays in contact with a surface while it moves from one place to another, a gripper shifts or seats its hold again in the air, or a gripper opens over the table or over open space instead of over the spot the thing is going to.
- **SOP rule broken:** Steps 1 to 5, each thing is lifted clear before it moves, held without a new grip in the air, and released over its destination.
- **Coaching note:** one thing per pick, lift before you carry, set it down before you take a new hold, and know where it lands before you open the gripper.

Violation: Placement shoved or fiddled with

- **Visible cue:** a gripper nudges, pats, or slides the peeled fruit or a banana segment into position after releasing it instead of lifting it clear and placing it again, or makes more than about two small corrections to settle a placement or the row.

- **SOP rule broken:** Steps 2.1 and 4, correct a bad placement by lifting the thing clear and placing it again, and each placement settles in one or two small corrections.
- **Coaching note:** lift it and lay it again rather than shoving it. If the first placement lands wide every time, tighten the approach rather than correcting after.

Violation: Check skipped

- **Visible cue:** a step's end state is plainly wrong on camera — skin left on the flesh, a fifth segment, two segments still joined, a gap in the row, pulp on the board — and the next step starts anyway with no look and no retry.
- **SOP rule broken:** Steps 1 to 5, each step's check is confirmed before the next step starts.
- **Coaching note:** stop and look at the end of every step. A check you skip is a violation the reviewer tags twice.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** the banana, the peel, a segment, the knife, or the cloth falls to the table or the floor short of its spot, the waste bin is struck and shifts, the row is swept off the plate, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the row broken, the knife off its rest, a peel on the board or the table, fruit or pulp on the board, the cloth off its rest, an arm away from home, or an arm still over the plate.
- **SOP rule broken:** Step 6, confirm the plate and the clear spots; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective banana:** bruised, split, or so ripe that the flesh breaks apart under a correct peel, or a banana stem that snaps off when the first peel strip is pulled. Replace it before the next episode.

- **Defective plate:** cracked, wet, or dirty when the episode begins. Replace it before the next episode.
- **Defective knife:** blunt enough that a correct cut crushes the flesh instead of cutting it, bent, or dirty when the episode begins. Replace it before the next episode.
- **Defective cloth:** already wet, torn, or shedding fibre onto the board. Replace it before the next episode.
- **Banana slides** under a correct stem hold because the board is wet or slick. Wipe and dry the board before the next episode.
- **Plate slides** on the table under a correct cut because the table is wet or slick. Dry the table before the next episode.
- **Board slides** on the table under a correct press because the table is wet or slick. Dry the table before the next episode.

Annotation subtasks (from SOP)

1. Peel the banana on the board, turning it between pulls
2. Move the peeled fruit to the plate and throw the banana peel away
3. Cut the peeled fruit into four banana segments on the plate
4. Arrange the banana segments into one row on the plate
5. Wipe the board and park the cloth
6. Confirm the plate and end the episode